Introduction to Artificial Intelligence and Machine Learning (CSN12601)

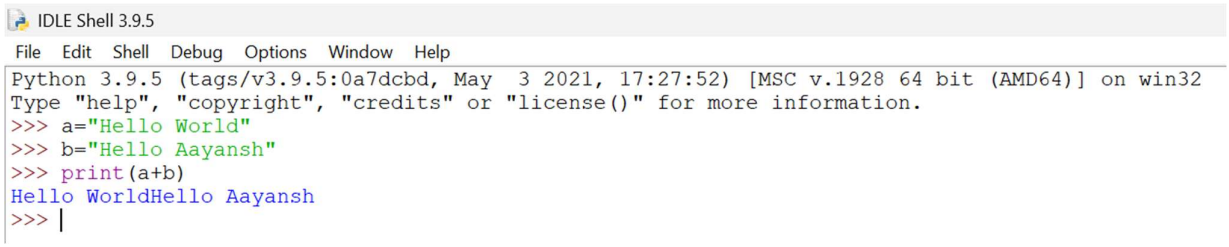
ASSIGNMENT-1

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Q1. Installation of Python and demonstration to run programming using Interpreter, IDLE, Command Prompt, and Google Colaboratory.

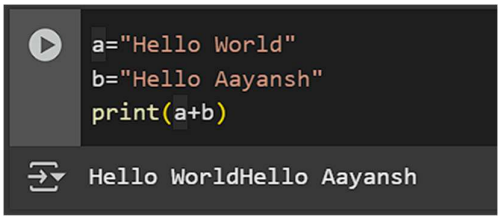
INTERPRETER :



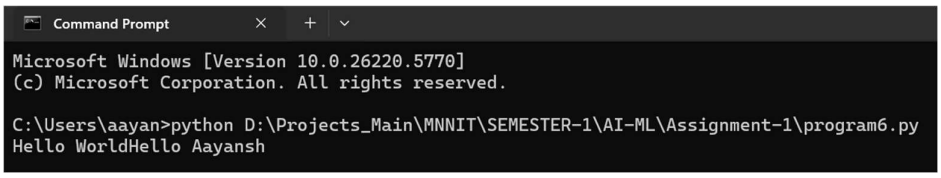
IDLE:



GOOGLE COLAB:



COMMAND PROMPT:



Q2. Write a program to demonstrate different number datatypes of python

A2. CODE :

```
a=25
b=25.75
c=True
print("Integer:",a,"\nType:",type(a))
print("Float:",b,"\nType:",type(b))
print("Boolean:",c, "\nType:",type(c))
```

OUTPUT:

```
Integer: 25
Type: <class 'int'>
Float: 25.75
Type: <class 'float'>
Boolean: True
Type: <class 'bool'>
```

Q3. Python Programs to demonstrate of various Type Conversion Functions.

A3. CODE :

```
a = 10
b = 25.75
c = "50"
print("Integer:", a, "Type:", type(a))
print("Float:", b, "Type:", type(b))
print("String:", c, "Type:", type(c))
print()
print("Integer to float:", float(a),"Type:",type(float(a)))
print("Float to integer:", int(b),"Type:",type(int(b)))
print("Integer to string:", str(a),"Type:",type(str(a)))
print("String to integer:", int(c),"Type:",type(int(c)))
```

OUTPUT:

```
Integer: 10 Type: <class 'int'>
Float: 25.75 Type: <class 'float'>
String: 50 Type: <class 'str'>

Integer to float : 10.0 Type: <class 'float'>
Float to integer : 25 Type: <class 'int'>
Integer to string : 10 Type: <class 'str'>
String to integer : 50 Type: <class 'int'>
```

Q4. Write a program to perform different arithmetic operations on numbers in python.

A4. CODE:

```
a = 15
b = 4
print("First Number (a):", a)
print("Second Number (b):", b)
print()
print("Addition (a+b):",a+b)
print("Subtraction (a-b):",a-b)
print("Multiplication (a*b):",a*b)
print("Division (a/b):",a/b)
print("Modulus (a%b):",a%b)
print("Exponent (a**b):",a**b)
print("Floor Division (a//b):",a//b)
```

OUTPUT:

```
First Number (a): 15
Second Number (b): 4

Addition (a+b): 19
Subtraction (a-b): 11
Multiplication (a*b): 60
Division (a/b): 3.75
Modulus (a%b): 3
Exponent (a**b): 50625
Floor Division (a//b): 3
```

Q5. Write a program that include operators like "and", "or", "not" and Bitwise operators

i. Bitwise and (&)
ii. Bitwise or (l)
iii. Bitwise ex-or (A)
iv. Bitwise complement (—)
v. Bitwise left shift Operator (<<), and Bitwise right shift Operator

A5. CODE :

```
x = True
y = False
print("x AND y:",x and y)
print("x OR y:", x or y)
print("NOT x:", not x)
print()
a = 10
b = 4
print("Bitwise Operators:")
print("a=",a,"Binary:",bin(a))
print("b=",b,"Binary:",bin(b))
print()
print("a&b (Bitwise AND):",a&b,"Binary:",bin(a&b))
print("a|b (Bitwise OR):",a|b,"Binary:",bin(a|b))
print("a^b (Bitwise XOR):",a^b,"Binary:",bin(a^b))
print("~a (Bitwise NOT):",~a,"Binary:",bin(~a))
print("a<<1 (Left Shift):",a<<1,"Binary:",bin(a<<1))
print("a>>1 (Right Shift):",a>>1,"Binary:",bin(a>>1))
```

OUTPUT :

```
x AND y: False
x OR y: True
NOT x: False

Bitwise Operators:
a= 10 Binary: 0b1010
b= 4 Binary: 0b100

a&b (Bitwise AND): 0 Binary: 0b0
a|b (Bitwise OR): 14 Binary: 0b1110
a^b (Bitwise XOR): 14 Binary: 0b1110
~a (Bitwise NOT): -11 Binary: -0b1011
a<<1 (Left Shift): 20 Binary: 0b10100
a>>1 (Right Shift): 5 Binary: 0b101
```

Q6. Write a program to create, concatenate and print a string and accessing substring from a given string.

A6. CODE:

```
a="Introduction to"
b="Artificial Intelligence and Machine Learning"
print("Concatenation (a+b): ",a+b)
print("Printing : ",b)
print("String slicing : ",b[0:23])
```

OUTPUT:

```
Concatenation (a+b):  Introduction toArtificial Intelligence and Machine Learning
Printing :  Artificial Intelligence and Machine Learning
String slicing :  Artificial Intelligence
```